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MATHEMATICS

STANDARD FOUR

National Assessment 2021
Answer Key and Worked Solutions

Subject	Mathematics
Code	04E
Level	Standard Four
Year	2021
Exam Board	NECTA
Exam	National Assessment
Type	Answer key and worked solutions
Questions	5 questions, each with 5 parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S4-2021.pdf

Question 1: Numbers and Place Value

1(a). Find the place value of the fourth digit from the right in 49,051.

Step 1: From the right: 1 is ones, 5 is tens, 0 is hundreds, 9 is thousands.

Step 2: The digit 9 is in the thousands place and represents $9 \times 1,000$.

Final Answer: Thousands place; its value is 9,000.

1(b). Write 32 in Roman numerals.

Step 1: $30 = XXX$ and $2 = II$.

Step 2: Combine the symbols.

Final Answer: XXXII.

1(c). Write 76,502 in expanded form.

Step 1: 7 is ten-thousands, 6 is thousands, 5 is hundreds, 0 is tens, 2 is ones.

Step 2: Add the non-zero place values.

Final Answer: $70,000 + 6,000 + 500 + 2$.

1(d). Write $2,000 + 900 + 70 + 5$ in short form.

Step 1: $2,000 + 900 = 2,900$.

Step 2: $2,900 + 70 + 5 = 2,975$.

Final Answer: 2,975.

1(e). Write 435 in words.

Step 1: $400 =$ four hundred.

Step 2: $35 =$ thirty-five.

Final Answer: Four hundred and thirty-five.

Section Summary: Read each digit's place and distinguish ordinary numbers from Roman numerals.

Question 2: Sequences and Comparison

2(a). Find the missing number: 50, 40, 30, __, 10.

Step 1: The sequence decreases by 10.

Step 2: $30 - 10 = 20$.

Final Answer: 20.

2(b). Arrange 23, 64, 32, 46, 29, 38, 54 from largest to smallest.

Step 1: Compare the tens digits first.

Step 2: Within the same tens group, compare the ones digits.

Final Answer: 64, 54, 46, 38, 32, 29, 23.

2(c). Find the next Roman numeral: V, XVI, XXVII, XXXVIII, __.

Step 1: Convert to 5, 16, 27, 38.

Step 2: Each term increases by 11, so $38 + 11 = 49$.

Step 3: $49 = 40 + 9 = XL + IX$.

Final Answer: XLIX.

2(d). The English paper lists heights 112 cm, 124 cm and 136 cm for Mrisho, Samwel and Rose respectively. Find Rose's height.

Step 1: The word 'respectively' matches the heights to the names in order.

Step 2: Rose is third, so her given height is 136 cm.

Final Answer: 136 cm.

Study Tip: This English version differs from the Kiswahili paper, which names Zalia and asks for Rose's next height.

2(e). Starting at 20, add 5 each time. Find the fifth number.

Step 1: Four increases are needed after the first term.

Step 2: $20 + 4 \times 5 = 40$.

Final Answer: 40.

Section Summary: Find the repeated change first; count the gaps between terms.

Question 3: Operations and Time

3(a). Add 14 hours 30 minutes and 8 hours 40 minutes.

Step 1: $30 + 40 = 70$ minutes = 1 hour 10 minutes.

Step 2: $14 + 8 + 1 = 23$ hours.

Final Answer: 23 hours 10 minutes.

3(b). Add 340 boys and 318 girls.

Step 1: $340 + 300 = 640$.

Step 2: $640 + 18 = 658$.

Final Answer: 658 pupils.

3(c). Subtract 4 hours 50 minutes from 6 hours 45 minutes.

Step 1: Borrow 1 hour: 6 h 45 min = 5 h 105 min.

Step 2: $105 - 50 = 55$ minutes; $5 - 4 = 1$ hour.

Final Answer: 1 hour 55 minutes.

3(d). Convert two hours into minutes.

Step 1: 1 hour = 60 minutes.

Step 2: $2 \times 60 = 120$.

Final Answer: 120 minutes.

3(e). Ronaldo played 40 minutes of a 90-minute match. How long did he not play?

Step 1: Subtract the time played from the full match time.

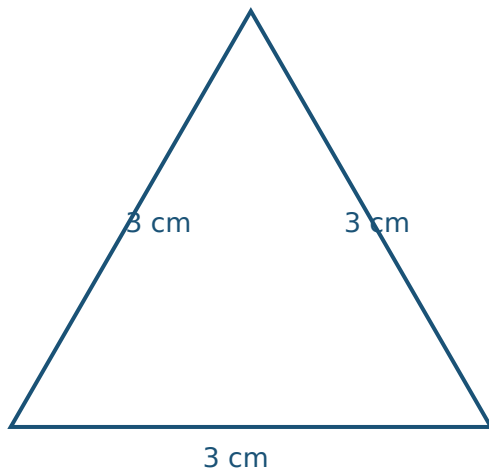
Step 2: $90 - 40 = 50$.

Final Answer: 50 minutes.

Section Summary: Keep the units consistent. Regroup every 60 minutes as one hour.

Question 4: Shapes and Measurement

4(a). Draw an equilateral triangle.



Step 1: Choose a side length, for example 3 cm, and draw a 3 cm base.

Step 2: Using a compass set to 3 cm, draw arcs from both endpoints and mark their intersection.

Step 3: Join the intersection to the endpoints; all three sides are 3 cm.

Final Answer: An equilateral triangle, with all three sides equal.

4(b). Find the perimeter of the rectangle with sides 8 cm and 5 cm.

Step 1: Opposite sides of a rectangle are equal.

Step 2: Add all four sides: $8 + 5 + 8 + 5$.

Final Answer: 26 cm.

4(c). An equilateral triangle has perimeter 57 cm. Find one side.

Step 1: There are three equal sides.

Step 2: Divide the perimeter by 3: $57 / 3 = 19$.

Final Answer: 19 cm.

4(d). Find the perimeter of a square with side 6 cm.

Step 1: A square has four equal sides.

Step 2: $4 \times 6 = 24$.

Final Answer: 24 cm.

4(e). A triangle has perimeter 45 cm and two sides 12 cm and 15 cm. Find the third side.

Step 1: Known sides: $12 + 15 = 27$ cm.

Step 2: Third side = $45 - 27$.

Final Answer: 18 cm.

Section Summary: Use the shape's sides, perimeter formula and metric units carefully.

Question 5: Reading Data

5(a). Find the total pupils in the pictograph: Dodoma has 3 symbols, Tanga 1.5, Iringa 2; one full symbol = 50,000 pupils.

Step 1: Dodoma: 150,000; Tanga: 75,000; Iringa: 100,000.

Step 2: Add $150,000 + 75,000 + 100,000$.

Final Answer: 325,000 pupils.

5(b). Find the pupils in Iringa and Tanga together.

Step 1: Iringa has $2 \times 50,000 = 100,000$ pupils.

Step 2: Tanga has $1.5 \times 50,000 = 75,000$ pupils.

Step 3: $100,000 + 75,000 = 175,000$.

Final Answer: 175,000 pupils.

5(c). Which region has the least pupils?

Step 1: Compare Dodoma 150,000, Tanga 75,000 and Iringa 100,000.

Step 2: The smallest count is 75,000.

Final Answer: Tanga.

5(d). What fraction of all pupils are in Tanga and Iringa together?

Step 1: Their combined count is 175,000; the total is 325,000.

Step 2: $175,000/325,000 = 175/325$.

Step 3: Divide numerator and denominator by 25.

Final Answer: 7/13.

5(e). Find the difference between the greatest and least regional pupil counts.

Step 1: The greatest count is Dodoma's 150,000; the least is Tanga's 75,000.

Step 2: $150,000 - 75,000 = 75,000$.

Final Answer: 75,000 pupils.

Section Summary: Read the key, count every symbol, and distinguish the year from the numerical value.